



Computer Vision Fundamentals

- Introduction
- Pixel Representation
- Spatial Resolution

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What is Computer Vision?

Computer Vision is an AI discipline that gives machines the ability to capture, process, analyze, and interpret visual data (images & video) to perform tasks that mimic human sight.



Face Recognition

Identify people in photos or live video with feature embeddings and matching.



Medical Imaging

Detect anomalies in X-rays, MRIs, and CT scans to assist diagnosis.



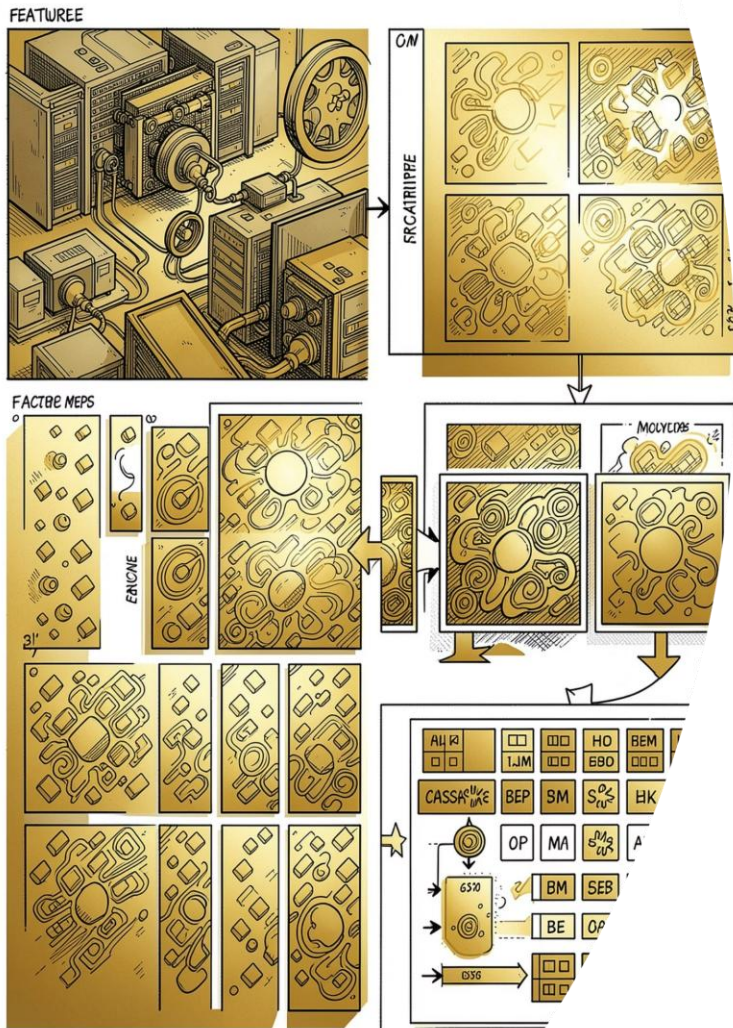
Autonomous Driving

Perceive lanes, vehicles, pedestrians for safe navigation decisions.



Robotics & Inspection

Automate quality control and guide robotic manipulation using vision.



Core Features & Levels of Computer Vision

Key Features

- Image acquisition & pre-processing
- Feature extraction (handcrafted & learned)
- Object detection & classification
- Segmentation and scene parsing
- Decision-making & tracking

Levels of Vision

Low-level: Capture, denoise, basic filtering (pixels → signals).

Mid-level: Segmentation, edges, feature descriptors, matching.

High-level: Object recognition, scene understanding, behavior inference.

□ Design systems by matching algorithmic complexity to the vision level required.

What Is a Pixel?

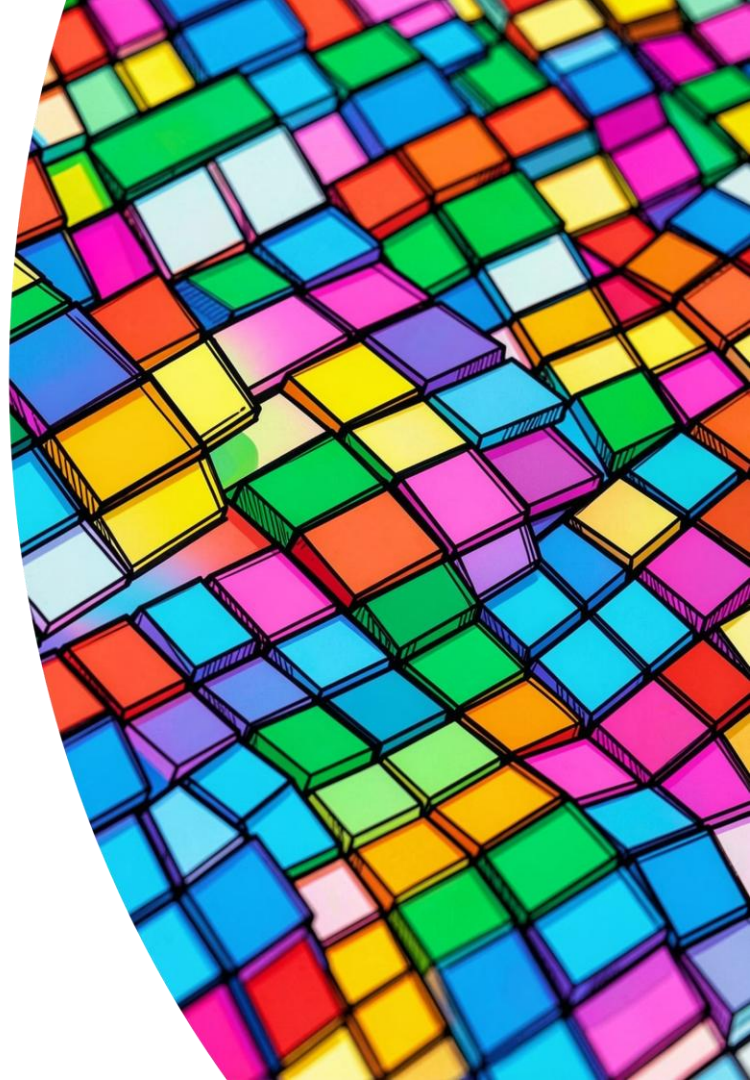
A pixel is the smallest addressable element of an image. It encodes brightness (grayscale) or color (RGB channels). Digital images are matrices of pixel values used as input to vision algorithms.

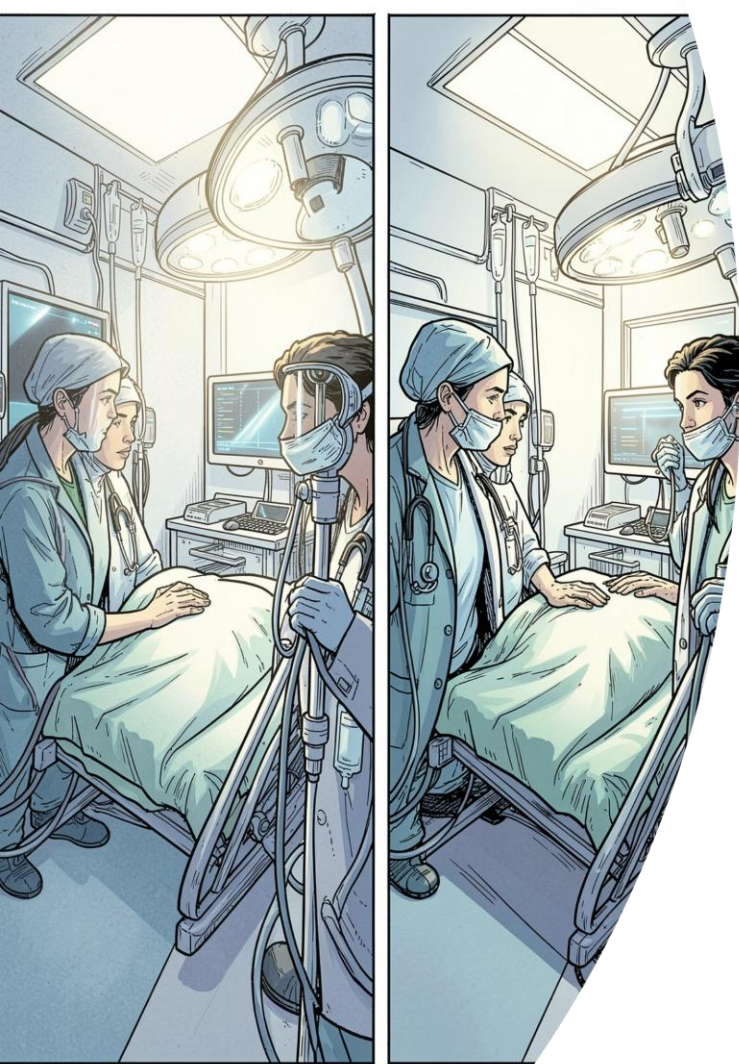
- **Types of Pixel-Based Images**

- Binary — two values (foreground/background)
- Grayscale — single intensity channel
- Color (RGB) — three channels per pixel

- **Why Pixels Matter**

- Form the numerical input for filters and neural nets
- Local pixel patterns drive feature extraction
- Sampling, quantization, and noise affect downstream accuracy





Common Challenges in Pixel-Based Processing

Real-world images suffer distortions that reduce recognition performance. Understanding these issues helps choose corrective techniques.

Noise

Random fluctuations (sensor, ISO) — use denoising filters or learned models.

Blur

Motion or defocus — compensated with deblurring or by capturing higher-frame-rate data.

Pixelation & Low Resolution

Insufficient sampling — consider super-resolution or multiscale features.

Compression Artifacts & Dead Pixels

Artifacts distort textures; use restoration or robust feature descriptors.

Pixel Representation in Real Applications

Pixel-level signals feed a wide range of tasks — accuracy at this stage often determines system reliability.

- **Facial recognition:** landmarks, embeddings derived from pixel patches.
- **Medical imaging:** intensity contrasts reveal pathology (tumor boundaries).
- **OCR:** pixel strokes become character shapes; binarization & morphological ops help.
- **Satellite imaging:** spectral pixels detect land use and changes over time.
- **Industrial inspection:** micro-level pixel defects flag manufacturing faults.



□ Better pixel capture and preprocessing → higher downstream recognition accuracy.

Spatial Resolution — Definition & Examples

Spatial resolution = number of pixels representing an image (width × height). Higher resolution captures finer detail but increases storage and compute.

01

Low Resolution (e.g., 256×256)

Faster processing, less detail — may miss small objects or textures.

03

High Resolution (e.g., 1024×1024+)

Captures small features—useful for medical, satellite, and inspection tasks; higher costs.

02

Medium Resolution (e.g., 512×512)

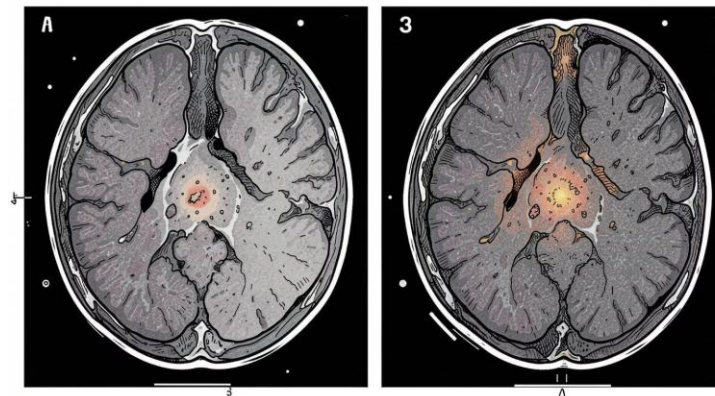
Balanced detail and compute for many detection/classification tasks.



Why Spatial Resolution Matters

- **Medical imaging:** detect sub-mm tumors with fine resolution.
- **Satellite imagery:** resolve roads, small buildings, vegetation changes.
- **Face recognition:** finer details improve landmark localization and identity verification.
- **Industrial inspection:** tiny defects become visible at higher resolutions.

Trade-offs: Storage $\leftrightarrow$ Compute $\leftrightarrow$ Latency. Choose an optimal resolution for the task and available resources.



Practical Recommendations & Solutions

Match preprocessing and resolution to the application. Combine classical filters with learned models to improve robustness.

Capture Better Data

Use appropriate sensors, controlled illumination, and calibration.

Choose Resolution Strategically

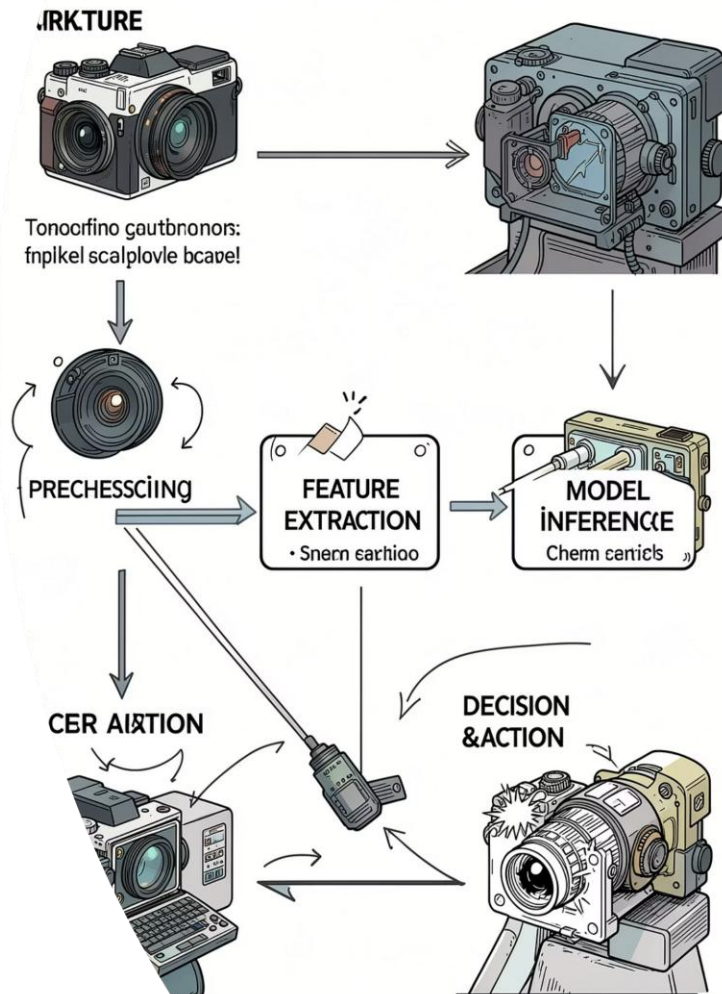
Balance task requirements against compute and storage constraints.

Preprocess Intelligently

Denoise, normalize, and correct distortions before feature extraction.

Use Multi-Scale & Learned Models

Combine pyramids, super-resolution, and deep networks for robust performance.



Summary & Next Steps

- Computer vision turns pixel arrays into actionable understanding.
- Pixel quality and spatial resolution are foundational to accuracy.
- Address noise, blur, and compression early via preprocessing and sensor choices.
- Select resolution and algorithms that fit your application constraints.



Further Reading

Intro CV textbooks, survey papers on deep learning for vision, and practical tutorials.



Hands-on Practice

Build small projects: image classification, detection, and super-resolution pipelines.



Questions?

Thank you — open for discussion.

